

Selected Answers to Review Problems

1. (a) e^6
(b) $-\frac{1}{2}$
(c) 0
(d) 2
2. (a) $e^{-x^2}(3x^2 - 2x^4)$
(b) $\frac{(1+x^2)\left(\frac{1}{2\sqrt{x}}\right) - \sqrt{x}(2x)}{(1+x^2)^2}$
(c) $(1+x^8)^x \left[\ln(1+x^8) + \frac{8x^8}{1+x^8} \right]$
3. $2 \times 2 \times 2$ meters
4. $-\frac{5}{4}$
5. $\frac{31!}{14!}$
6. $-\frac{1}{2} \sum_{k=1}^{\infty} \frac{x^{4k}}{k}$
7. (a) -3
(b) -9
8. (a) $\frac{1}{2}(x^2+1)\ln(x^2+1) - \frac{1}{2}(x^2+1) + C$
(b) $\frac{1}{2}(x+1)^2 - 2(x+1) + \ln|x+1| + C$
(c) $\frac{1}{5}e^{5x} - \frac{1}{3}e^{-3x} + C$
(d) $-\frac{1}{2}(x^2+1)^{-1} + C$
9. (a) $\frac{1}{2}$
(b) $\frac{27}{2} + \ln 4$
(c) $\frac{1}{3} \ln 9$

10. (a) Converges
 (b) Diverges
 (c) Diverges
 (d) Converges
11. (a) $-3x^2e^{-x^6}$
 (b) $\sqrt{(\ln x)^4 + x^3} \left(\frac{1}{x} \right) + \int_1^{\ln x} \frac{3x^2}{2\sqrt{t^4 + x^3}} dt$
 (c) $\ln(1 + \sin^{10} x) \cos x - \ln(1 + x^5) \frac{1}{2\sqrt{x}}$
12. 10
13. $2 \ln 10 + 3 \left(2 \ln 10 + \frac{4}{10} \right) - \frac{24}{10}$
14. (a) $\int_0^1 \int_{-\sqrt{y}}^{\sqrt{y}} f(x, y) dx dy$
 (b) $\int_0^2 \int_{3x}^6 f(x, y) dy dx$
15. (a) $\frac{1}{20}$
 (b) $\frac{4\pi}{3}$
16. (a) $y(t) = \ln \left(\frac{1}{2}e^{2t} + \frac{1}{e} - \frac{1}{2} \right)$
 (b) $y(t) = \frac{5}{3} - \frac{11}{3}e^{-3t}$
 (c) $y(t) = e^{t^2} - e^{t^2} \int_0^t e^{-s^2-s^3} ds$
17. (a) $y(t) = C_1t^2 + C_2t^3$
 (b) $y(t) = C_1e^{6t} + C_2e^{-6t}$
 (c) $y(t) = C_1e^{2t} + C_2te^{2t}$
18. (a) $-12 - 2\sqrt{5}$
 (b) 0
19. 3, -5, 6
20. 0
21. $(I - A)^{-1} = I + A + A^2 + \cdots + A^{n-1}$